

Predictors as Markets

One maker, makers in parallel, markets in series

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Working draft v0.2 · August 28, 2026

Abstract

Many learning procedures are cost-function markets by identity rather than analogy: mirror descent trades against a convex-cost maker, Bayesian model averaging is a market of Kelly bettors with wealths as posterior weights, least squares is the precision-weighted merge of one quadratic maker per observation, and a proximal step is the optimal response to a maker charging the penalty. This paper asks the converse, which predictors are markets, and answers in three movements. For one maker, the apparent obstruction, non-convexity, does not obstruct coherence: charges telescope for any cost, no-arbitrage is a chord condition, and rational flow trades the biconjugate, landing on the contact set so that non-convexity costs expressiveness (holes in the book, multimodal quotes) rather than soundness; a proportional fee is exactly a bid-ask spread, by conjugation, and a fee of at least the chord excursion restores no-arbitrage. For makers in parallel, combining fee-bearing makers is an infimal convolution solved by one monotone clearing-price root-find with lasso-sparse fills; the aggregate supply curve is a consolidated limit order book, each maker may quote its own fee with routing as the discipline, and a deep quadratic co-quoter is Moreau smoothing. For markets in series, one market per chain-rule factor opens when its conditioning information freezes and settles in cascade: alternating a model step with a market step is the Kalman filter, market messages on a tree compute exact posterior marginals, and on loopy graphs the pathology of belief propagation materializes as arbitrage, with arbitrageurs as the loop correction. The two compositions are the two operations of the inference semiring, and Gaussian exactness throughout has one source, the family on which sum-product and min-sum agree. A predictor is a market when its arbitrage depth is finite relative to affordable friction; a market is the incentive closure of a predictor.

1 The correspondences, and the question

A cost-function market maker posts a potential C over inventory and charges $C(q + s) - C(q)$ for a fill s ; prices are gradients and, when C is convex, the standard theory applies (Hanson 2003; Abernethy et al. 2013). A run of published identities makes many learning procedures markets in this literal sense. Trading against such a maker is follow-the-regularized-leader with the cost as conjugate regularizer (Chen and Vaughan 2010; Abernethy et al. 2013), with trader wealth as the learning rate (Fronçillo et al. 2012). A market of Kelly bettors performs Bayesian model averaging with wealths as posterior weights (Beygelzimer et al. 2012), and equilibrium prices of utility-maximizing agents implement mixtures and products of experts (Pennock and Wellman 1997; Storkey 2011; Storkey et al. 2012; Barbu and Lay 2012). Ordinary least squares is a market of data points: each observation a quadratic maker quoting its value with capital equal to its precision, the estimate their merge, since merging makers is the infimal convolution

of their costs and liquidity adds (Bhaskara et al. 2023; Barrieu and El Karoui 2005). A proximal step is a trade against a fee-bearing maker: the proximal operator of $f|\cdot|$ is the soft-threshold, which is the optimal response to a proportional fee (Lemma 5 below), and the prox of any convex g is the response to a maker charging g . The two canonical penalties are then the two market primitives: ridge is a zero-quoting participant with capital λ , lasso is a fee of λ , and the theorem that regularization is robustness to data perturbation (El Ghaoui and Lebrete 1997; Xu et al. 2009) acquires a market reading in which the adversary’s budget is priced rather than assumed.

The question is the converse: which predictors are markets, and what does the market form demand? The paper proceeds in three movements. Sections 2–4 treat one maker and show the suspect requirement, convexity, is weaker and more informative than supposed. Sections 5–7 treat makers in parallel: fees, routing, the order book, and the frictions that repair incoherence. Sections 8–11 treat markets in series: the chain rule as architecture, filtering, message passing, and loops. Section 12 states the characterization. Throughout, a scalar security settles at $\varphi(\omega) = \omega \in [-1, 1]$; vector statements substitute the convex hull of payoff vectors. Reference implementations and numerical theorem tests accompany the paper in the mechanisms repository.

2 Coherence is a chord condition

Proposition 1 (no-arbitrage without convexity). *For a path-independent maker with cost C (no convexity assumed), there is no outcome-independent profitable fill from any state if and only if every chord slope of C lies in the convex hull of payoffs: for all q, s ,*

$$C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega) s \quad (\text{scalar: } C \text{ is 1-Lipschitz}).$$

Proof. A fill s from state q has sure profit $\min_{\omega} \varphi(\omega) s - [C(q + s) - C(q)]$; positivity for some pair (q, s) is exactly the failure of the displayed inequality, and the inequality for all pairs says every chord slope is supported by the hull. ■

Two remarks. Cycling was never the exploit: the charge telescopes over any closed path whatever C is, so round trips are refunds; the exploit, when the chord condition fails, is the accumulation of sure-profit net positions at states where quotes exit the hull. And convexity enters the standard axiomatics not through no-arbitrage but through *information incorporation* (the marginal-cost monotonicity condition of Abernethy et al. (2013)); dropping that axiom while keeping the chord condition leaves a coherent non-convex maker, exhibited numerically in the companion repository. Arbitrage theory without convexity is developed in a different formalism by Lépinette and Tran (2017).

3 The market trades the biconjugate

Let $\hat{C}$ denote the lower convex envelope of a chord-coherent cost C (its biconjugate on the line (Rockafellar 1970)) and $g = C - \hat{C} \geq 0$ the gap.

Proposition 2 (contact and pass-through). *A myopic risk-neutral trader with believed mean μ in the hull, facing the maker at state q : (i) has optimal fills landing on the contact set $\{C = \hat{C}\}$, and (ii) earns maximal expected profit*

$$\Pi_C(q, \mu) = \Pi_{\hat{C}}(q, \mu) + g(q),$$

the envelope profit plus the gap at the starting state.

Proof. $\sup_s \mu s - [C(q + s) - C(q)] = \sup_x [\mu x - C(x)] - \mu q + C(q)$. Since $C \geq \hat{C}$ with equality on the contact set, and a maximizer of an affine function minus C is a point where an affine minorant touches C , hence touches $\hat{C}$, the supremum equals $\sup_x [\mu x - \hat{C}(x)]$ and is attained on the contact set. Adding and subtracting $\hat{C}(q)$ gives (ii). ■

The maker behaves observationally like its convex envelope. Concave stretches are unquotable intermediate states, holes in the book that rational flow jumps across; whoever lands inside one (noise) overpays the gap at the landing state, and by (ii) the next rational trader recoups it, so the maker is a conduit keeping envelope differences over any rational-to-rational span, and off-contact states are transient. The trade-set analogue on the CFMM side is the canonical concave trading function: an arbitrary invariant is behaviorally equivalent to a concave one (Angeris et al. 2024), with the limits of concavification mapped by Frongillo et al. (2024). Read as beliefs, a hole is a multimodal quote: the maker takes prices on either side of the gap and refuses every price inside, jumping discontinuously as flow accumulates, the book-side shape of the bimodal implied densities documented around binary events (Melick and Thomas 1997; Clark and Amen 2017).

4 Frictions price the remaining failure

The failure mode surviving §2 is chord slopes exiting the hull.

Proposition 3 (fees buy bounded incoherence). *If chord slopes exit the hull by at most ε per unit, i.e. $C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega)s - \varepsilon|s|$, then a proportional fee $f \geq \varepsilon$ restores no-arbitrage.*

Proof. Sure profit with the fee is $\min_{\omega} \varphi(\omega)s - \Delta C - f|s| \leq (\varepsilon - f)|s| \leq 0$. ■

This is the mechanism-level counterpart of a theorem of mathematical finance: arbitrarily small proportional transaction costs restore no-arbitrage for price processes that are arbitrageable frictionlessly, with a consistent price system inside the spread as certificate (Guasoni 2006; Guasoni et al. 2010). In prediction markets the precedent is fees sized to expected arbitrage profit restoring bounded loss under privacy noise (Cummings et al. 2016; Frongillo and Waggoner 2018). The minimum viable spread of a cost is its arbitrage depth, and a venue quoting a wide spread to cover a wild landscape is coherent but uninformative in proportion: the market prices the model’s incoherence as uncertainty.

5 A linear fee is a bid-ask spread

Why a fee at all? A cost-function maker’s charge telescopes, so round trips are free: the maker earns nothing on uninformed flow and cannot recover adverse-selection losses from volume. That no state-dependent cost can charge a round trip, and that a path-dependent volume charge repairs it, is due to Othman and Sandholm (2012); the linear charge $f|s|$ is their volume levy in its simplest convex form, chosen because it conjugates in closed form. Write $g_q(s) = C(q + s) - C(q)$ for a convex C and $m = C'(q)$ for the marginal price.

Lemma 4 (fee–spread duality). *Let $T_f(x) = \text{sign}(x) \max(|x| - f, 0)$ denote the soft-threshold. Then the fee-bearing cost $\tilde{C}_q = g_q + f|\cdot|$ has conjugate*

$$\tilde{C}_q^*(p) = g_q^*(m + T_f(p - m)),$$

zero if and only if $|p - m| \leq f$.

Proof. The conjugate of a sum of closed proper convex functions with overlapping relative interiors of domains is the infimal convolution of the conjugates (Rockafellar 1970, Thm. 16.4), and the conjugate of $f|\cdot|$ is the indicator of $[-f, f]$. Hence $\tilde{C}_q^*(p) = \min_{|u| \leq f} g_q^*(p - u)$, the minimum of the convex function g_q^* over $[p - f, p + f]$. Since $g_q^* \geq 0$ with equality exactly at m , the minimum is attained at the projection of m onto the interval, which is $m + T_f(p - m)$. ■

The maker quotes $\text{ask} = m + f$ and $\text{bid} = m - f$ and is untouchable in between: a proportional fee is not like a spread, it is one, the same duality by which a proportional transaction cost confines the pricing functional to the bid-ask band (Jouini and Kallal 1995). And because conjugation reads only biconjugates, the lemma survives non-convexity verbatim with the envelope in place of the cost:

Lemma 5 (the envelope form). *For any chord-coherent C with envelope $\hat{C}$, the fee-bearing maker's no-trade band at state q is the interval of beliefs within f of the envelope's marginal price, non-empty exactly on the contact set; at off-contact states every belief in the hull yields profit at least $g(q)$.*

Proof. By Proposition 2, profit at belief μ is the envelope profit plus $g(q)$; with the fee the envelope profit is the soft-thresholded conjugate of Lemma 4, zero on the band around $\hat{C}'(q)$. On the contact set $g(q) = 0$ and the band survives; off it the additive $g(q) > 0$ leaves no zero. ■

6 Makers in parallel: routing, the order book, Moreau

Let makers $i = 1..n$ hold inventories q_i with convex costs C_i , liquidities of their choosing, and fees f_i of their choosing.

Lemma 6 (combination). *Let $\tilde{C} = \tilde{C}_1 \square \dots \square \tilde{C}_n$ and define each maker's supply*

$$s_i(p) = \begin{cases} (C'_i)^{-1}(p - f_i) - q_i, & p \geq \text{ask}_i, \\ 0, & \text{bid}_i < p < \text{ask}_i, \\ (C'_i)^{-1}(p + f_i) - q_i, & p \leq \text{bid}_i, \end{cases}$$

each non-decreasing in p . Fix a demand Δ for which a clearing price p^ with $\sum_i s_i(p^*) = \Delta$ exists. Then: (i) $\tilde{C}^* = \sum_i \tilde{C}_i^*$, a sum of soft-thresholded profit functions; (ii) the split $s_i = s_i(p^*)$ attains $\tilde{C}(\Delta)$, and any optimal split satisfies $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$ for $s_i \neq 0$ and $|p^* - m_i| \leq f_i$ for $s_i = 0$; (iii) the split is sparse: every maker whose quote band strictly contains p^* trades exactly zero.*

Proof. (i) is the conjugate-sum identity applied to the convolution (Rockafellar 1970). For (ii), the split is feasible by choice of p^* , and $p^* \in \partial \tilde{C}_i(s_i(p^*))$ for every i : when $s_i(p^*) \neq 0$ the subgradient is $C'_i(q_i + s_i) + f_i \text{sign}(s_i) = p^*$, and when $s_i(p^*) = 0$ it is the interval $[m_i - f_i, m_i + f_i] \ni p^*$. A common multiplier certifying every coordinate is exactly the optimality condition for $\min\{\sum_i \tilde{C}_i(s_i) : \sum_i s_i = \Delta\}$. (iii) restates the zero branch. ■

The computation is a scalar monotone root-find whatever n is, and the fee costs nothing beyond a horizontal shift of each supply curve. The $|s|$ terms are an ℓ_1 penalty, so sparsity arrives for the same reason it does in the lasso, and for the same reason proportional transaction costs

produce no-trade regions and sparse portfolios (Olivares-Nadal and DeMiguel 2018). A small trade routes entirely to the tightest quote; a growing trade pushes that maker's fee-adjusted marginal price through the next band and spills over, consuming makers in fee order.

Corollary 7 (zero fees). *With $f_i \equiv 0$ the convolution reduces to the fee-free merge: conjugate regularisers add, and for a perspective family $C_b(q) = b C_1(q/b)$ liquidity adds, $C_{b_1} \square C_{b_2} = C_{b_1+b_2}$ (Bhaskara et al. 2023).*

Corollary 8 (the order book). *The aggregate supply $S(p) = \sum_i s_i(p)$ is non-decreasing, identically zero on $(\max_i \text{bid}_i, \min_i \text{ask}_i)$ when that interval is non-empty, flat wherever every maker's band covers p , and smooth and strictly increasing wherever some maker is in the money. Read as a market: best bid and ask are the tightest quotes, depth at each price is the sum of the active makers' closed-form supplies, and large orders walk the levels. The aggregate of linear-fee makers is a consolidated limit order book, and in producer-theory terms Lemma 6 is Marshall's horizontal summation of firm supply curves (Marshall 1890; Mas-Colell et al. 1995) with the reversibility of share production patched by the fee. The economics of the book assembled from competing liquidity suppliers is Glosten (1994), with convergence of strategic schedules in Biais et al. (2000).*

Proposition 9 (a deep co-quoter is Moreau smoothing). *Merging a chord-coherent (possibly non-convex) maker with a quadratic co-quoter of liquidity λ yields the venue with cost the Moreau envelope $e_\lambda C(x) = \min_y C(y) + (x - y)^2/(2\lambda)$, whose arbitrage depth decreases to zero as $\lambda \rightarrow \infty$; generically (tilted gaps) the merged venue is exactly convex at finite depth, while a symmetric double well retains a concave kink of magnitude $O(1/\lambda)$ at every depth, priced by a fee of the same order.*

Proof sketch. The merge is infimal convolution, and inf-convolution with the quadratic is the Moreau envelope (Rockafellar 1970). $e_\lambda C$ is a minimum of convex branches $y \mapsto C(y) + (\cdot - y)^2/(2\lambda)$; non-convexity survives only at branch crossings, where the slope jump is the branch-minimizer separation over λ , hence $O(1/\lambda)$, vanishing or merging entirely when the crossing is eliminated by tilt. ■

The two repairs are the two market primitives again: friction (ℓ_1 , the fee) and participant depth (ℓ_2 , capital), lasso and ridge.

7 Self-set fees, stabilizers, adaptivity

Nothing requires the fees to be administered. Each maker may quote its own f_i : a quote inside the aggregate spread earns nothing, a quote too tight is picked off by informed flow, and the undercutting happens inside the same minimisation that clears the trade, since the inf in the convolution is a minimum over quotes. The surviving fee is the competitive adverse-selection charge of classical microstructure (Glosten and Milgrom 1985; Biais et al. 2000), reached through routing rather than a dealer game.

Optimization already pays these frictions: a proximal step charges $\|\Delta\theta\|^2/(2\eta)$ per move, a trust region is an infinite fee outside a band, weight decay is a zero-quoting participant. In market language the stabilizers that make non-convex training behave are the frictions that make a non-convex venue non-exploitable. Path-dependent optimizers correspond to makers whose quotes depend on flow history, the adaptive-liquidity territory where path independence is deliberately traded away: no maker combines path independence, translation invariance and liquidity sensitivity (Othman et al. 2013), the adaptive class is axiomatized by Li and Wortman Vaughan (2013) with the homogeneous-risk-measure characterization in Othman and Sandholm (2011), no trade-history maker achieves every desideratum at once (Abernethy et al.

2014), and liquidity selection itself can be run as online learning (Nueve et al. 2026; Nueve and Waggoner 2025).

8 Two operators, one semiring

Sections 5–7 composed makers on one quantity. A model is a factorization,

$$P(\text{everything}) = \prod_{\text{nodes}} P(x_i \mid \text{parents}(x_i)),$$

and the remaining operator runs one market per factor, each pricing its conditional given what is upstream. That there are exactly two operators is not an accident of taste. Message-passing inference is an algorithm over a commutative semiring (Aji and McEliece 2000): one operation combines evidence about a variable (the product), one moves evidence between variables (the sum), and sum-product, max-product and min-sum are the one algorithm over different semirings. The market algebra is that pair: parallel merge is the product, densities multiplying as precisions add, and serial propagation is the sum. The parallel operation wears two costumes, addition of conjugates and infimal convolution of costs, because inf-convolution is convolution in the min-plus semiring and the Legendre transform is its Laplace transform (Litvinov 2005): the convex machinery of this paper is the tropical shadow of the probability calculus. Exactness in everything Gaussian below has one source: log-quadratics are the family on which the sum-product and min-sum semirings coincide, so a market whose traders optimize agrees with an inference engine that integrates; off the family they diverge by the Laplace-approximation gap, which is §12’s open question.

9 One market per factor

The locality that makes the factorization tradable is Hanson’s modularity: in a combinatorial LMSR a bet on $A \mid B$ moves that conditional and provably nothing else, uniquely among market scoring rules (Hanson 2007, 2003). Hanson runs one joint market over the product space; the substance of pricing it is probabilistic inference, which is why exact pricing is #P-hard (Chen, Fortnow, et al. 2008), why tournament markets price by Bayes-net inference (Chen, Goel, et al. 2008), why a deployed combinatorial market ran its price and asset updates on the junction-tree algorithm (Sun et al. 2012), and why approximate designs price over the marginal polytope (Dudík et al. 2012; Dudík et al. 2021). Securities structured by a Bayes-net factorization appear in Pennock and Wellman (2000), with the trades that preserve the structure characterized by Xia and Pennock (2011).

The serial architecture takes the factors as separate venues rather than one joint book. Market i opens on $P(x_i \mid \text{parents})$ when its conditioning information freezes, conditions on upstream quotes while upstream is live, and re-references when upstream settles; settlements cascade through the DAG like dataflow, and the schedule of markets is an unrolling of the graph. Observable nodes carry settled markets; a latent node either settles through its observable footprint, a market on a hidden state being a market on a functional of future observables scored through the model, or remains indicated but unsettled, priced by whoever warehouses the model risk.

10 The market step is the Kalman update

Take the linear-Gaussian chain $x_{t+1} = ax_t + \varepsilon_t$, $y_t = x_t + \eta_t$, and alternate two steps: a *model step*, propagating the current belief through the dynamics in mean-precision form, and a *market step*, merging the propagated belief with an observation maker quoting y_t whose capital is the observation precision.

Proposition 10. *The alternation is the Kalman filter (Kalman 1960): after each observation the market state equals the filtered mean and variance exactly.*

Proof. The model step is the standard prediction of mean and variance. Merging quadratic makers adds precisions and precision-weights means (the fusion form of Corollary 7), which is the information-form measurement update; the covariance-form update follows by the usual algebra. ■

Prediction is computation; correction is a market; capital decides how hard the data pulls. The nearest published object is the Bayesian market maker of Brahma et al. (2012), whose trade update is a scalar Gaussian measurement update with covariance inflation for jumps; the filtering reading, and any alternation with a dynamic model, appear unstated in the literature.

11 Trees and loops

On a Gaussian chain with an observation at every node, the posterior at an interior node is the merge of three sources: the forward message (the filter of §10 run up to the node), the local observation, and the backward message (downstream observations pulled back through the dynamics, each pull-back a reparametrization and each combination a precision addition).

Proposition 11. *All three messages are market operations, and the merged posterior equals the brute-force conditioning of the joint at every node: Gaussian belief propagation (Pearl 1988) with trades as messages, exact on trees.*

The repository verifies the identity to ten digits. The contrast with the deployed combinatorial engines is architectural: there the junction tree is the pricing algorithm inside one joint market (Sun et al. 2012); here the messages pass between venues, and the graph’s edges are market boundaries.

On loopy graphs belief propagation double-counts and its fixed points drift. The market version of the pathology is sharper and self-punishing.

Proposition 12. *Assemble pairwise correlation quotes around a cycle into a quote matrix P with unit diagonal. The quotes admit a joint distribution iff $P \succeq 0$. Otherwise, with w the eigenvector of a negative eigenvalue λ , the bundle with weights $ww^\top$ has price $w^\top Pw = \lambda < 0$ and payoff $(w^\top x)^2 \geq 0$: a sure profit of at least $|\lambda|$ per unit.*

This is the second-moment coherence condition, prices of products must form a PSD matrix (d’Aspremont 2005), read as cycle consistency. The lineage of “arbitrageurs enforce coherence” is long and must be owned: coherence is no-arbitrage (Nau and McCardle 1991); Pennock and Wellman (1996) built arbitrageur agents enforcing the additivity identities of a Bayes-net economy, with equilibrium prices equal to the network’s probabilities and distributed bidding as distributed inference; the combinatorial literature removes the arbitrage algorithmically over the marginal polytope (Kroer et al. 2016); and Polymarket’s arbitrageurs have been measured doing the enforcement for eight figures (Saguiillo et al. 2025). What the loop reading adds is the object being corrected: the inconsistency that degrades loopy belief propagation is, in a market, not a numerical nuisance but free money, and the flow that harvests it pushes the quotes back

toward the cone. Whether the corrected fixed point is the true marginal, a Bethe-like surrogate, or something the liquidity profile selects is open.

12 The characterization, and open problems

A predictor is a market when its cost has finite arbitrage depth relative to affordable friction; convexity is the $f = 0$ special case. Within the coherent class, non-convexity spends expressiveness, the contact set in place of a full quoting range, and spends nothing else against rational flow. What the market form adds to a predictor is not new estimates but three things the estimation literature obtains otherwise: tuning constants become prices set by competition rather than formulas, composition becomes an algebra, parallel and serial as the two semiring operations, and the procedure becomes robust to strategic data, since every source pays to move the state. A market is the incentive closure of a predictor; a predictor is a market with the strategyproofing stripped out.

Open problems, in rough order of tractability:

Which inference algorithm is a market? A market of optimizing traders natively computes min-plus, so off the Gaussian family it prices the max-product posterior rather than the sum-product marginal, the discrepancy being the Laplace-approximation gap. Risk aversion is an exponential tilt, suggesting it interpolates between the two semirings; whether a risk-averse trading population prices the marginal, the mode, or a temperature in between would say precisely which inference algorithm a market runs.

The equilibrium fee. Section 7 argues discipline, not equilibrium. In a sparse-signal flow model, does the Bertrand-equilibrium fee reproduce the universal-threshold rate of the shrinkage literature, making “the right amount of regularization is the adverse-selection cost of the data source” a proposition rather than a slogan?

Loopy fixed points. Under a concrete flow model, does arbitrage-corrected quoting converge, and to what?

Arbitrage depth as a capacity measure. The minimum viable spread of a trained landscape is computable; whether it tracks quantities learning theory names (sharpness, mode connectivity) is open.

Books around binary events. The multimodal-quote reading of book holes (§3) is testable against limit-order data around court rulings and FDA decisions, where bimodal implied densities are documented (Clark and Amen 2017).

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